Hiring Assessment - AI-Generated song detection: Attribution of Synthetic Music

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1 Introduction

This report details the design, implementation, and evaluation of an AI-Original Pairwise Similarity system, developed to fulfill the requirements of the synthetic audio attribution assignment. As generative AI models for music (such as Suno and Udio) become increasingly sophisticated, the ability to trace synthetic derivatives back to their human-made training data or prompts is a critical challenge.

The primary objective of this project is to detect and quantify the relationship between an original human-made track (Track A) and a potential AI-generated counterpart (Track B). Moving beyond simple binary classification, this system focuses on relational modeling to determine if Track B is a synthetic derivative, variation, or "hallucinated" generation based on Track A. The assignment specifically requires the system to handle variations in tempo, arrangement, and spectral artifacts across variable-length audio.

To achieve this, the solution employs a Multi-Task Siamese Network. By fusing classical acoustic features, high-order artifact signatures, and deep semantic embeddings, the architecture evaluates both the musical relationship and the presence of synthetic generation artifacts to output a definitive, continuous Attribution Score.

2 Main Idea and Solution Architecture

The core of the proposed solution is a Multi-Task Siamese Network [14]. Audio attribution cannot be solved by a simple binary classifier because the goal is relational: determining if Track B was derived from Track A. The network processes two audio inputs simultaneously through a shared Track Encoder to extract deep representations, which are then evaluated by three output heads (Similarity, AI Detection, Attribution) with a dedicated LyricEncoder module feeding into the Attribution Head:

- **Similarity Head**: Computes the organic, musical relationship between the tracks. This head evaluates shared acoustic and semantic properties to establish a baseline of relational overlap, regardless of whether the audio is real or synthetic.
- **AI Detection Head (AI Index)**: Acts as a per-track classifier to identify synthetic vocoder and diffusion artifacts. This produces an "AI Index," an independent probability score indicating the presence of generative footprints in a given track.
- **Attribution Head**: Synthesizes the outputs of the previous heads to output a continuous score (0.0 to 1.0) representing the likelihood of derivation. This serves as the final determination of whether the AI generation explicitly mimics or stems from the original human track.

To handle variable-length audio, the model employs a sliding-window chunking strategy. The audio is split into 10-second segments, processed through a Transformer-based ChunkEncoder, and aggregated using Attention Pooling. This design explicitly eliminates "window bias"—ensuring the model evaluates the entire track rather than just the intro—and allows it to focus on specific segments where hallucinated variations or artifacts are most prominent.

3 Datasets and Pair Construction

- **SONICS Dataset**: The primary source for large-scale synthetic songs [1], containing tracks from various generators such as Chirp-v3.5 and Udio-120s. It provides critical pair types for relational modeling:
  - *Lyric Pairs (Positive)*: Pairs an original human-made track with an AI-generated counterpart utilizing identical lyrics. This teaches the model to identify "prompt-hijacked" derivatives where the melody is hallucinated but the semantic core remains identical.
  - *Sibling Pairs (Positive)*: Two AI tracks generated from the same text prompt, teaching the model to distinguish broad stylistic similarity from true audio derivation.
  - *AI-to-AI Negatives*: Pairs tracks from the same generator with different content ("hard negatives") and tracks from competing generators (cross-generator negatives) to decouple model signatures from attribution logic.
  - *Real-to-AI Negatives*: Pairs unrelated AI-generated tracks with human-made tracks from the real pool. This ensures the detection of an AI footprint does not automatically result in a high attribution score unless melodic or semantic derivation is also present.
- **FakeMusicCaps Dataset**: A diverse collection of multi-model text-to-music clips [2]. Positive pairs are created from songs that come from the same YouTubeID, and negatives from samples with different IDs. They are also used for cross-generator and cross-topic negative sampling to aggressively train the AI Detection Head on high-variance spectral artifacts.
- **MIPPIA (SMP) Dataset**: A dataset of real human covers and plagiarized tracks [3]. This is vital for teaching the model what organic, musical similarity looks like in the total absence of synthetic footprints.
- **Echoes Dataset**: A benchmarking dataset containing original Free Music Archive (FMA) tracks paired with their direct AI-generated derivatives [5]. Crucially, since Echoes was not mentioned in the assignment, it is strictly excluded from training, and serves as an Out-of-Distribution (OOD) testing ground. This provides the ultimate sanity check—proving the network learns fundamental attribution concepts rather than memorizing training distributions.

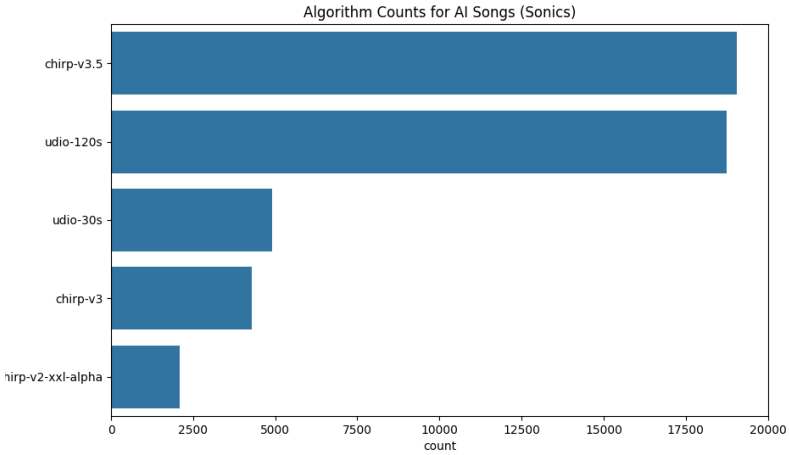


Figure 1: Distribution of AI generation algorithms in the SONICS dataset. Chirp-v3.5 and Udio-120s dominate the synthetic song distribution.

3.1 Splitting Strategy and Data Leakage Prevention

Training a Siamese network requires extreme care to prevent data leakage. If pairs are split randomly, an individual track might appear in both the training and validation sets, falsely inflating accuracy. Conversely, if individual tracks are randomly split 85/15, the probability of both tracks in a pair landing in the validation set is only 2.25%, leading to catastrophic "pair destruction". To solve this, the pipeline implements a rigorous track-level deterministic split based on source groupings rather than isolated files. For example, in the SONICS sibling dataset, tracks are grouped by a shared `group_id` representing a single generation prompt. In the MIPPIA dataset, covers are grouped by a shared `pair_number`. In the FakeMusicCaps dataset, tracks are grouped by their source YouTube ID, ensuring that same-prompt positive pairs (same YouTube ID, different generators) go entirely into one split rather than straddling both. The splitting function partitions these group identifiers (85% Training, 15% Validation) before any audio paths are resolved or paired. By ensuring an entire sibling group or cover family goes exclusively into either the training or validation set, this approach mathematically guarantees zero data leakage while preserving valid relational pairs.

4 Feature Construction and Selection Rationale

A robust attribution system requires a multi-faceted representation of the audio signal. Relying solely on one type of feature risks the model either overfitting to a specific generative architecture or failing to recognize abstract musical similarity. Therefore, the system constructs a fused representation comprising Classical, High-Order, and Deep Semantic features.

4.1 Classical Acoustic Features

Rationale: To establish a baseline of the musical texture, rhythm, and harmonic content. **Features Used**: Mel-frequency cepstral coefficients (MFCCs), Mel-spectrograms, Chroma, Tonnetz, and Spectral Contrast—yielding a fixed 432-dimensional vector per track.
Source Justification: Historically, audio detection and classification methods have relied on specific statistical features inherent to audio signals—most notably MFCCs—to build a foundational mathematical representation of the sound [10]. In our architecture, these classical features allow the Siamese network's Similarity Head to assess the broad strokes of melodic and harmonic alignment between two tracks.

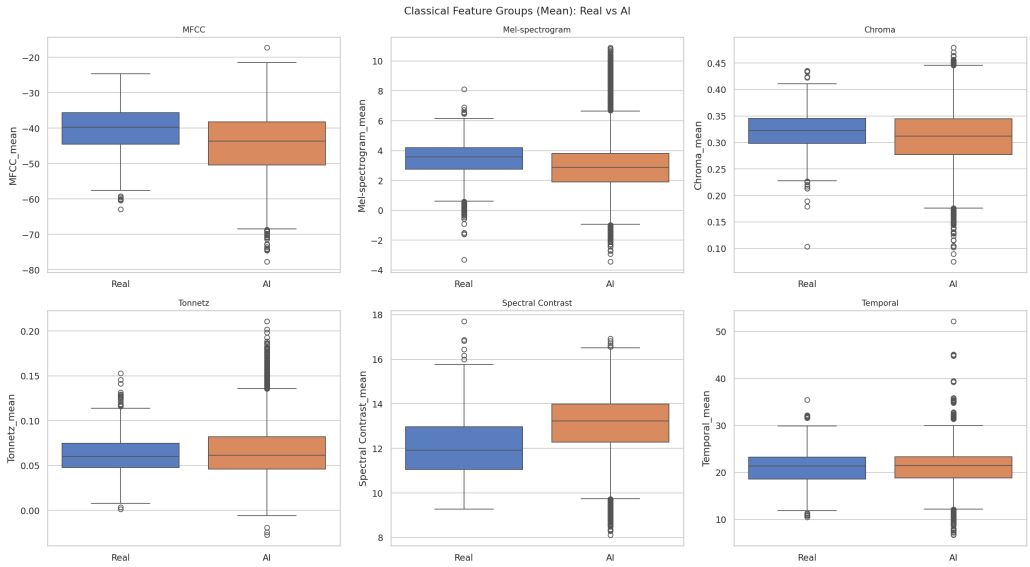


Figure 2: Classical feature group means for Real vs AI tracks. AI tracks show elevated Mel-spectrogram and spectral contrast means, reflecting the different timbral balance of neural vocoders.

4.2 High-Order Spectral Artifacts & The "Fakeprint"

Rationale: Modern generative audio networks (whether auto-regressive, diffusion, or GAN-based) typically utilize neural vocoders as a final step to synthesize audio waveforms from intermediate temporal-frequency representations. While these vocoders produce highly realistic sound, they often struggle with phase continuity and leave behind distinct structural anomalies.
Features Used: Phase Deviation, Instantaneous Frequency (IF) Stability, Harmonic-to-Noise Ratio (HNR), and a custom 897-dimensional "Fakeprint" (measuring Peak Regularity and Maximum Peak-to-Background ratios in the 1–8 kHz band) based on [8].
Source Justification: Classical AI detection features such as Phase Deviation, IF Stability, and HNR have been extensively validated as robust indicators of synthetic generation [12]. Furthermore, Fourier analysis proves that generative deconvolution architectures inherently produce regular spectral peaks, often referred to as "checkerboard" artifacts [8]. By isolating high-frequency peak regularity and phase inconsistencies, our AI Detection Head can fingerprint these artifacts without being confused by the underlying musical composition.

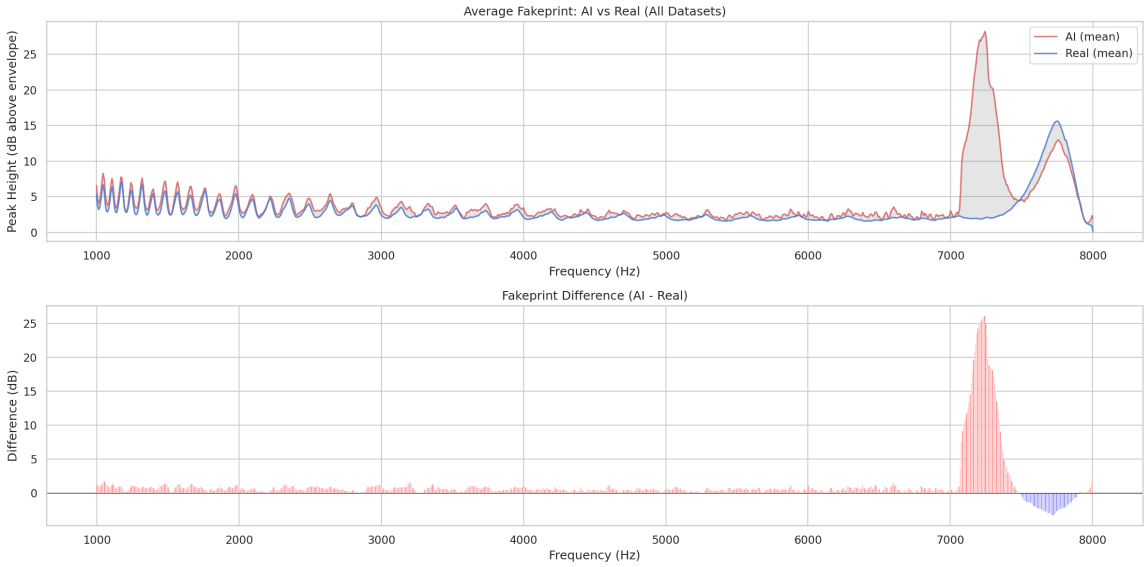


Figure 3: Average Fakeprint: AI vs Real across all datasets. Top: the 1–8 kHz spectrum shows a distinct AI spike near 7.2 kHz absent in real recordings. Bottom: the difference plot confirms the artifact is concentrated at this frequency.

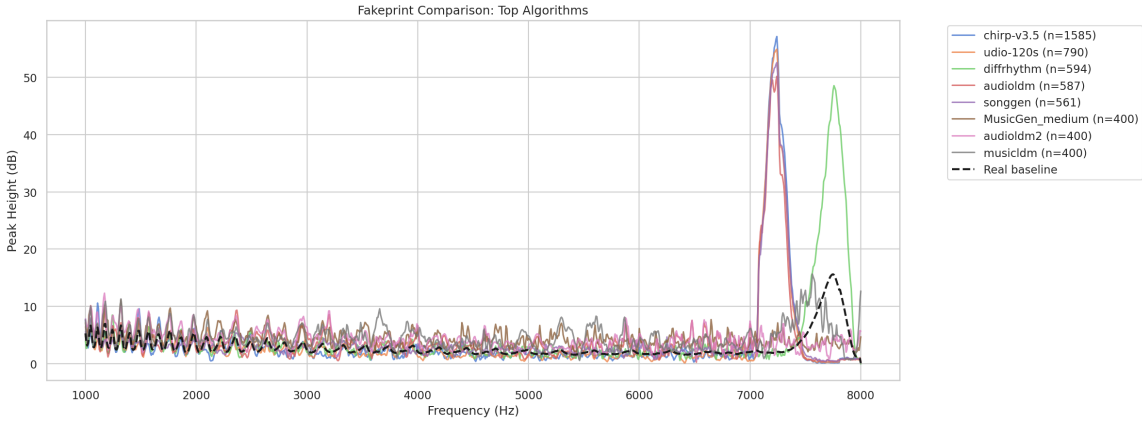


Figure 4: Fakeprint comparison across top AI generation algorithms. Each generator produces a characteristic spike pattern, enabling generator-level identification. The real baseline (dashed) is flat across the band.

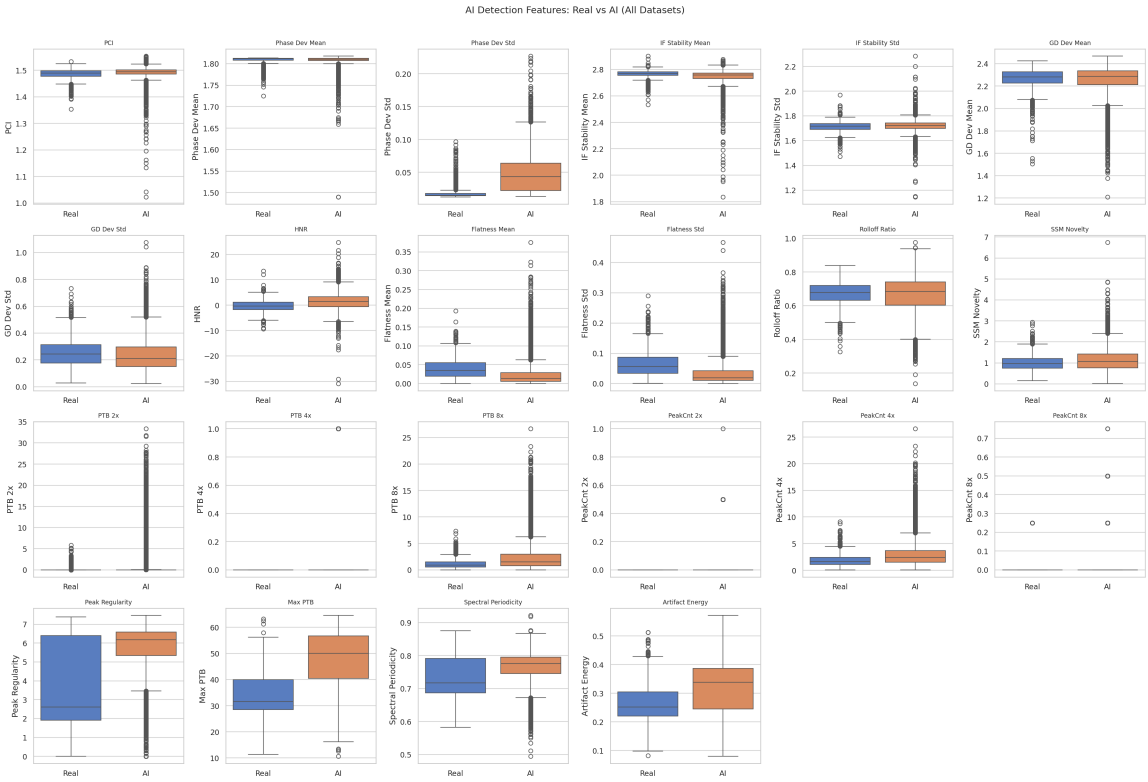


Figure 5: AI detection feature distributions (22 dimensions) for Real vs AI tracks across all datasets. Key discriminators include HNR, Phase Continuity Index, Spectral Periodicity, Peak Regularity, and Artifact Energy.

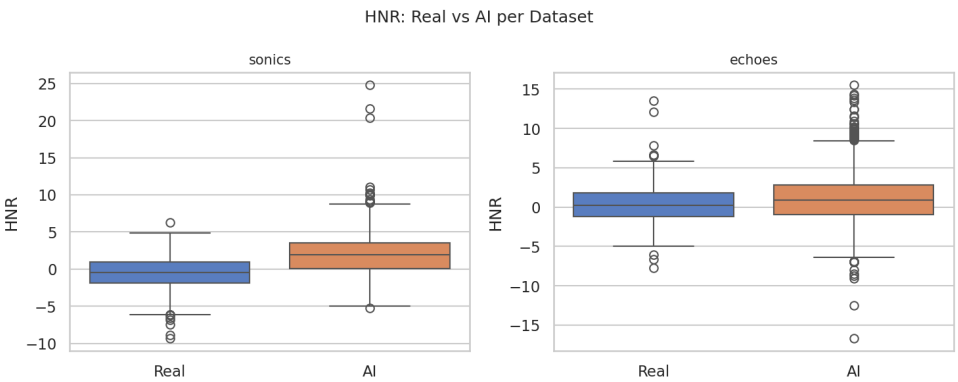


Figure 6: Harmonic-to-Noise Ratio (HNR) per dataset. AI-generated tracks in SONICS show consistently elevated HNR due to unnaturally clean harmonics produced by neural vocoders.

4.3 Deep Semantic & Relational Embeddings

Rationale: Mathematical features excel at detecting exact clones or artifacts, but they struggle to capture the complex semantic meaning of a song. To understand deep musical similarity, the architecture leverages foundation models. **Features Used**:

- **MERT** [9]: A self-supervised acoustic representation specifically tuned for music. By utilizing the 330M parameter model, the system extracts 1024-dimensional vectors that capture pitch, beat, and instrumentation.
- **CLAP** [13]: Connects language and audio by projecting them into a joint multimodal

space using contrastive learning. Because CLAP learns audio concepts through natural language supervision, it provides exceptional ability to identify relational and stylistic similarities across genres.

4.4 Lyric Embeddings (Whisper + SBERT)

Rationale: AI music generators are often used to generate entirely new melodies using identical lyrics—a form of "prompt hijacking."
Design Choice: By transcribing the audio via Whisper [11] and encoding the text into a 384-dimensional vector via SBERT [15], the model gains a secondary validation modality. Unlike a simple scalar cosine similarity, the system encodes a full pairwise interaction vector via a dedicated `LyricEncoder` module, projecting the concatenation of $[\mathbf{e}_a, \mathbf{e}_b, |\mathbf{e}_a - \mathbf{e}_b|, \mathbf{e}_a \odot \mathbf{e}_b]$ into a 64-dimensional feature that the Attribution Head can learn from.

4.5 Temporal Processing and Segmentation

Rationale: To capture the rhythmic and structural progression of the tracks, temporal features must be extracted over sequential, localized windows. **Design Choice**: Currently, the system utilizes fixed 10-second windows with 5-second overlap to roughly approximate 4-bar musical phrases. However, recognizing that rigid windowing can disrupt organic musical phrasing, future iterations aim to extend this logic to non-constant, structure-aware segment sizes. By dynamically aligning boundaries to actual musical bars rather than arbitrary fixed limits, the system would better capture underlying compositional similarity [3, 4].

5 Exploratory Data Analysis

Before model training, an extensive feature-level analysis was conducted to validate the chosen feature engineering and understand the statistical properties of the datasets. Key findings are summarized below.

5.1 Feature Set Discriminability

A Random Forest classifier trained independently on each feature group yields AUC > 0.95 for real-vs-AI separation. The combined feature set reaches near-perfect AUC. Among classical features, Mel-spectrogram is the single most important group, followed by MFCCs and temporal features.

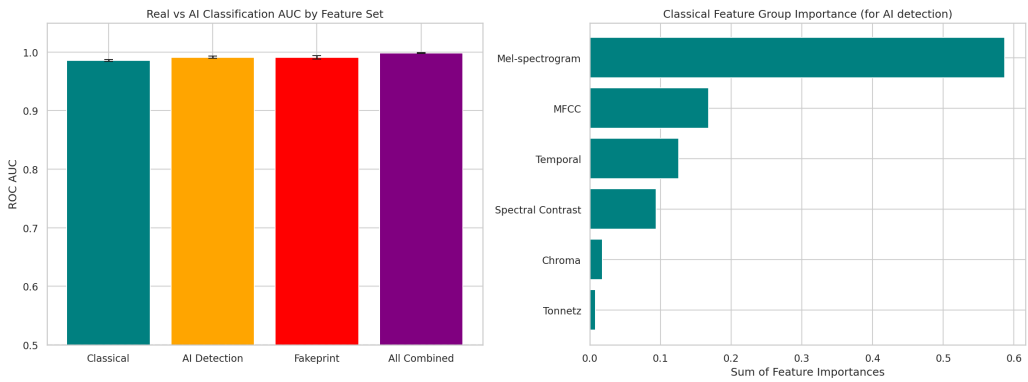


Figure 7: Real vs AI classification AUC by feature set (left) and classical feature group importance (right).

5.2 AI Detection Features Per Algorithm

AI detection features vary systematically by generator algorithm. Chirp-v3.5 and Udio-120s cluster tightly on Phase Continuity Index; diffrhythm shows the highest spectral periodicity and artifact energy.

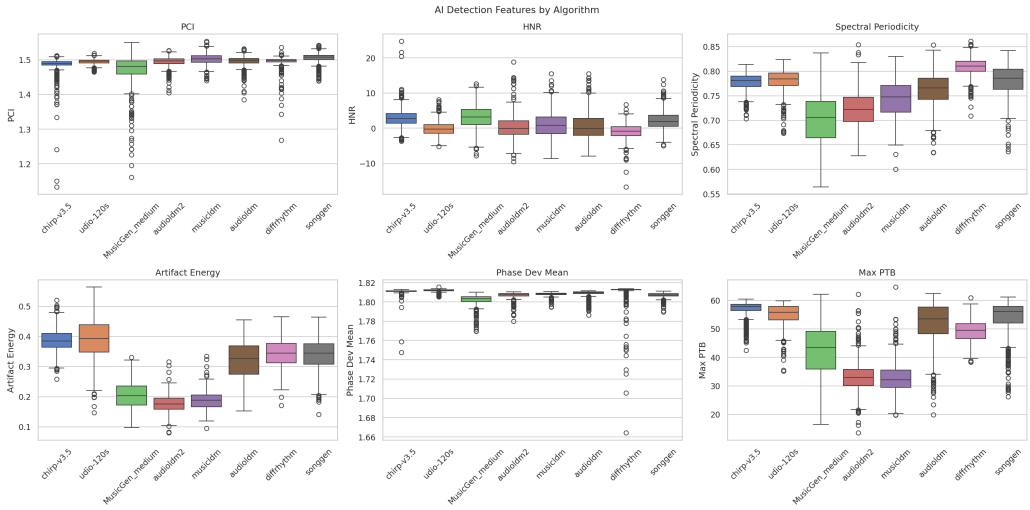


Figure 8: AI detection features by algorithm. Each generator occupies a distinct region of the feature space, supporting generator-level attribution.

5.3 t-SNE and Cross-Dataset Analysis

A t-SNE projection of the 22-dimensional AI detection features confirms strong real/AI separation and reveals that different datasets occupy distinct regions of the feature space.

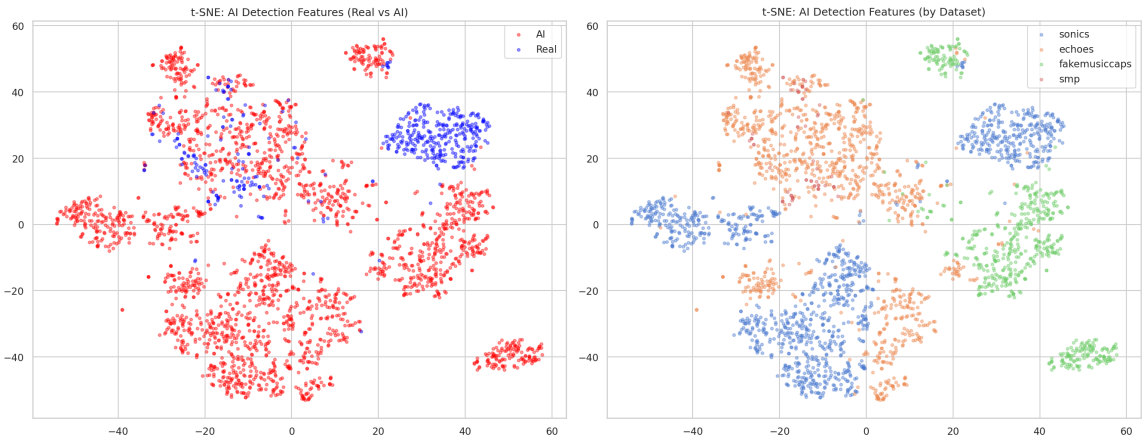


Figure 9: t-SNE projection of AI detection features. Left: Real (blue) vs AI (red) separation. Right: per-dataset clustering, showing distributional shift between SONICS, FakeMusicCaps, Echoes, and MIPPIA.

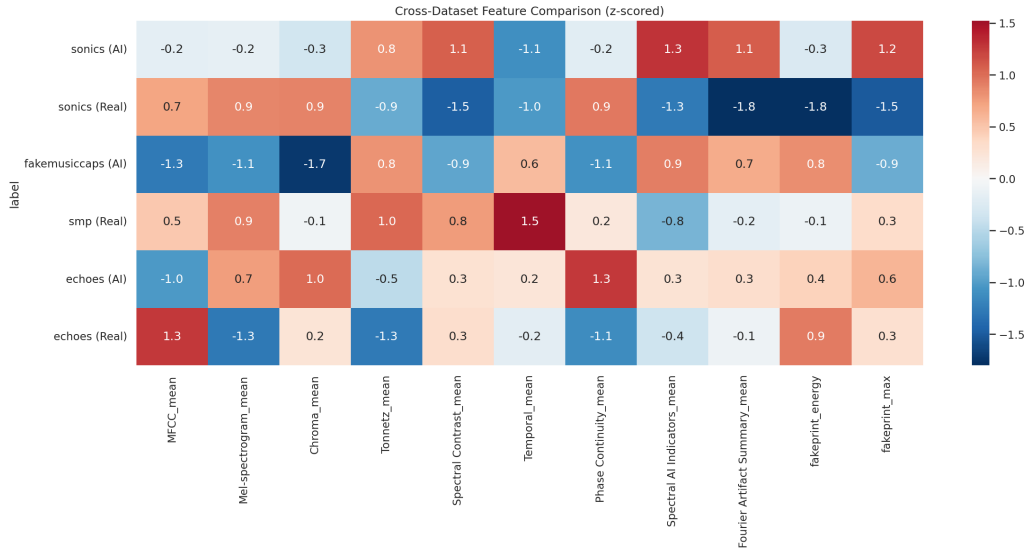


Figure 10: Cross-dataset feature comparison (z-scored). FakeMusicCaps (AI) shows low MFCCs/Mel means; MIPPIA shows high spectral contrast. The systematic differences motivate multi-dataset training.

5.4 Pair-Level Similarity

SONICS lyric pairs show high classical cosine similarity (∼0.99). MIPPIA pairs show near-identical similarity across relation types (plagiarism, remake, doubt).

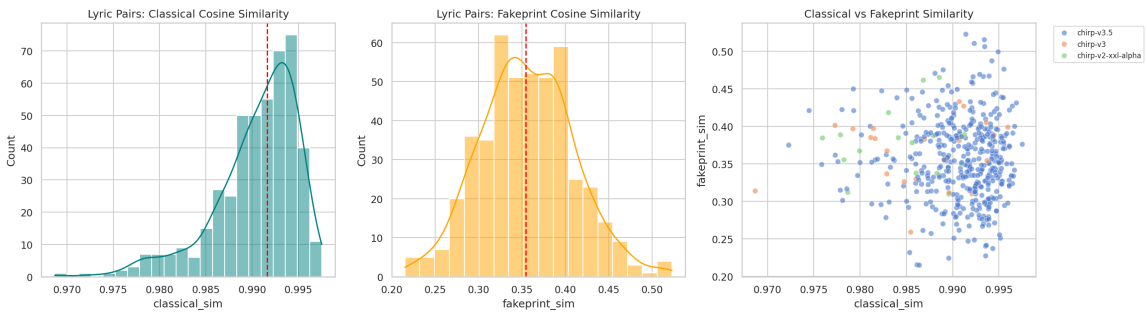


Figure 11: SONICS lyric pairs: classical cosine similarity (left), Fakeprint cosine similarity (center), and their scatter plot (right).

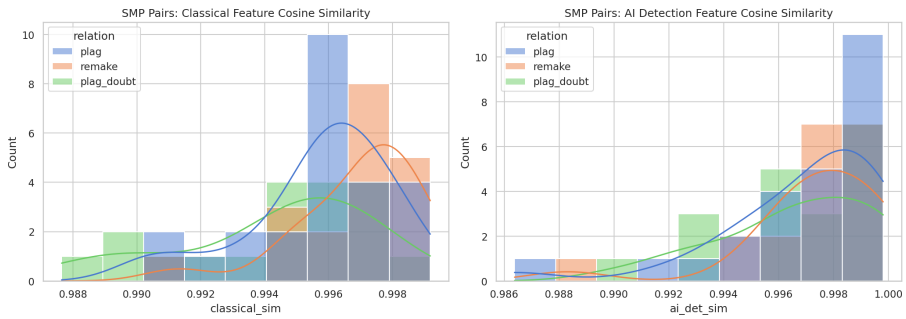


Figure 12: MIPPIA dataset pair similarity. Cover and plagiarism pairs show near-identical classical and AI detection similarity distributions, confirming consistent labelling.

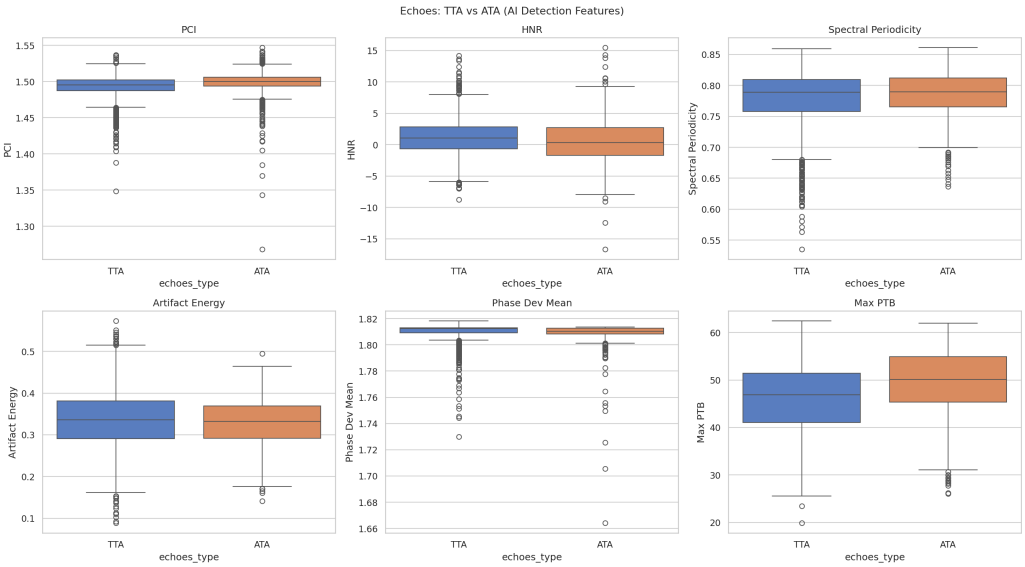


Figure 13: Echoes dataset: AI detection features for TTA (text-to-audio) vs ATA (audio-to-audio) generated tracks. ATA tracks show slightly elevated HNR and Max PTB, consistent with audio conditioning preserving more of the source harmonic structure.

6 Architecture Design

6.1 The Siamese Framework

The core model utilizes a Siamese (twin) network topology. Track A and Track B are passed through identical, weight-sharing sub-networks (Track Encoders) to project them into a shared latent space. By sharing weights, the Siamese network guarantees that if Track A and Track B have similar acoustic and semantic structures, their resulting latent vectors will be close in the multi-dimensional space, allowing for accurate relative comparison [6].

6.2 Temporal Modeling: ChunkEncoder and Attention Pooling

Audio tracks inherently vary in duration. To standardize this, the audio is divided into 10-second chunks with 5-second overlap. At training time, up to 12 chunks are randomly sampled (without replacement) from each track, with their temporal order preserved, to standardise sequence length across the batch and provide lightweight data augmentation. The extracted classical features for these chunks are processed sequentially by a 2-layer Transformer-based ChunkEncoder [7], followed by a learned Attention Pooling layer.

- **Overcoming Window Bias**: Many baseline models truncate audio to the first 30 seconds. This introduces severe "window bias," where a model might miss a plagiarized chorus or a synthetic bridge occurring later in the track.
- **Dynamic Focus**: The Attention Pooling mechanism assigns dynamic weights to different chunks. If an AI model faithfully copies the intro but produces an artifact-heavy section at the 2-minute mark, the attention layer can heavily weight that specific chunk.

6.3 Track Encoder and Embedding Fusion

MERT (1024-d) and CLAP (512-d) embeddings are projected to the model hidden dimension and fused with the chunk embedding through a learned gate:

$$\mathbf{h} = g \cdot \mathbf{h}_{\text{chunk}} + (1 - g) \cdot \mathbf{h}_{\text{fuse}}$$

where $g = \sigma(W[\mathbf{h}_{\text{chunk}}; \mathbf{h}_{\text{emb}}])$. This gating mechanism allows the model to adapt its reliance on classical versus semantic representations per sample.

6.4 Multi-Task Learning Heads

Once the Track Encoders reduce the variable-length sequences into fixed-size vectors, the representations are routed through four specialized heads:

- **Similarity Head**: Takes $[\mathbf{a}, \mathbf{b}, |\mathbf{a} - \mathbf{b}|, \mathbf{a} \odot \mathbf{b}]$ to compute a musical similarity logit.
- **AI Detection Head**: Operates individually on each track; concatenates the track embedding with the 22-d AI detection features and 897-d fakeprint to produce an AI index logit.
- **Lyric Encoder**: Projects $[\mathbf{e}_a, \mathbf{e}_b, |\mathbf{e}_a - \mathbf{e}_b|, \mathbf{e}_a \odot \mathbf{e}_b]$ from the 384-d SBERT lyric embeddings into a 64-d feature vector.
- **Attribution Head**: Synthesizes the similarity logit, both AI index logits, both track embeddings, and the lyric feature vector to produce the final attribution score.

6.5 Loss Function

The multi-task loss is:

$$\mathcal{L} = \lambda_{\text{sim}}\mathcal{L}_{\text{sim}} + \lambda_{\text{ai}}\mathcal{L}_{\text{ai}} + \lambda_{\text{attr}}\mathcal{L}_{\text{attr}} + \lambda_{\text{cons}}\mathcal{L}_{\text{cons}}$$

All task losses use binary cross-entropy with label smoothing ($\epsilon = 0.05$). Attribution positives are up-weighted $4\times$ to address class imbalance. The consistency regularizer $\mathcal{L}_{\text{cons}} = \text{ReLU}(p_{\text{attr}} - p_{\text{sim}})$ enforces the constraint that attribution cannot exceed similarity—if two tracks are not musically related, one cannot be attributed as a derivative of the other.

7 Similarity vs Attribution: A Critical Distinction

A central design challenge in this system is that *musical similarity and attribution are not the same thing*, and conflating them is the most common failure mode in naive approaches.

Similarity answers: "Are these two tracks musically related?" It is high whenever two tracks share melodic, harmonic, or timbral content—regardless of how or by whom they were created. A human cover of a pop song is highly similar to the original. Two AI tracks generated from the same style prompt are highly similar to each other. An AI remix of a folk melody is highly similar to its source.

Attribution answers: "Was one of these tracks derived from the other?" It is specifically high when Track B is a *synthetic derivative* of Track A—when an AI model has used Track A (or its prompt, lyrics, or melodic structure) as a direct basis for generation.

The distinction matters in several concrete cases:

- **Sibling pairs** (two AI tracks from the same prompt): similarity ≈ 0.8, attribution $= 0$. They are related but neither is derived from the other.
- **Real covers** (MIPPIA dataset): similarity $= 1.0$, attribution $= 1.0$. One was created as a deliberate derivative of the other.
- **Unrelated AI tracks**: similarity ≈ 0, attribution $= 0$. Both happen to be AI-generated but share no content.

- **Lyric pairs** (real track + AI fake from same lyrics): similarity = 1.0, AI Index B ≈ 1, attribution = 1.0. This is the canonical positive case.

The architecture encodes this distinction explicitly. The **Similarity Head** evaluates musical content overlap between tracks, while the **Attribution Head** receives the similarity logit *alongside* both AI index logits, both track embeddings, and the lyric interaction vector. It must learn that high similarity is a *necessary but not sufficient* condition for attribution.

This is further enforced by the **consistency regularizer**: $\mathcal{L}_{\text{cons}} = \text{ReLU}(p_{\text{attr}} - p_{\text{sim}})$, which penalizes any prediction where attribution exceeds similarity. The model is explicitly trained to never claim derivation without underlying musical relationship.

8 Evaluation

8.1 Training Data Statistics

The pair construction pipeline produced the following training and validation split:

Pair Source	Train	Val
SONICS lyric positives	392	70
SONICS sibling positives	850	150
MIPPIA (cover/plagiarism) positives	55	8
FakeMusicCaps same-prompt positives	127	23
SONICS same-generator negatives	566	100
SONICS cross-generator negatives	566	100
Fake vs unrelated real negatives	1,004	175
FakeMusicCaps cross-generator negatives	510	90
Cross-dataset negatives (SONICS $\leftrightarrow$ FMC)	127	23

Table 1: Training and validation pair counts by source. Splits are track-level: no individual audio file appears in both sets.

The negative-to-positive ratio of $\approx$ 2:1 is intentional. The same-generator negatives (two tracks from the same generator, different content) are the hardest negatives: they share fakeprint signatures, requiring the model to rely on musical content rather than artifact identity to assign low attribution.

8.2 Ablation Study

The model was evaluated across four feature configurations:

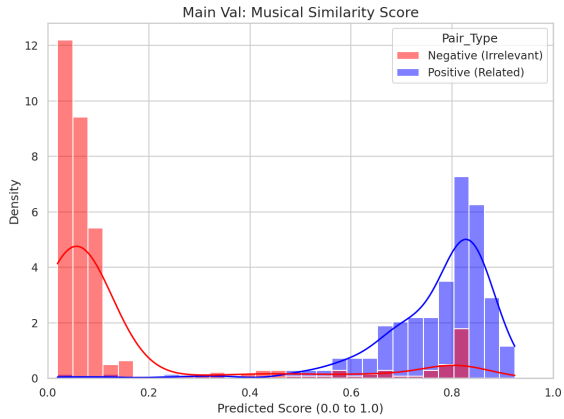
Feature Set	Val Loss	Sim Acc	AI-A Acc	AI-B Acc	Attr Acc
basic	2.017	**0.784**	0.791	0.900	0.841
embedding	2.146	0.494	0.569	**0.995**	**0.850**
advanced	1.815	0.771	**0.999**	0.902	0.823
mix (advanced + lyrics)	**1.741**	0.765	0.997	0.902	0.832

Table 2: Ablation results. Accuracy is rounded threshold accuracy at 0.5. "basic": classical features only. "embedding": MERT + CLAP only. "advanced": classical + fakeprint + AI detection + embeddings. "mix": advanced + LyricEncoder.

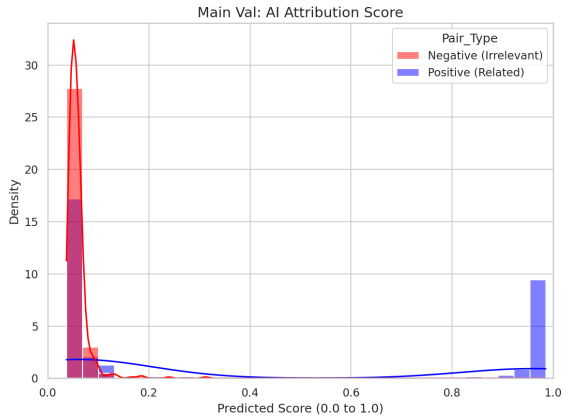
Key observations:

- **Advanced significantly improves AI detection**. Adding fakeprint and AI detection features (22 + 897 dims) pushes AI-A accuracy to 99.9% and lowers validation loss (2.017 → 1.815), though basic acoustic features remain highly competitive for raw similarity and attribution.
- **Embedding-only model has poor AI-A detection** (56.9%). MERT/CLAP embeddings encode musical content but not artifact signatures—without the fakeprint, Track A (the real track) cannot be reliably identified as human-made.
- **Lyrics improve attribution** (0.823 → 0.832) at a small cost to similarity (0.771 → 0.765). The LyricEncoder provides additional signal for the attribution head on lyric pairs that the acoustic features alone cannot distinguish.

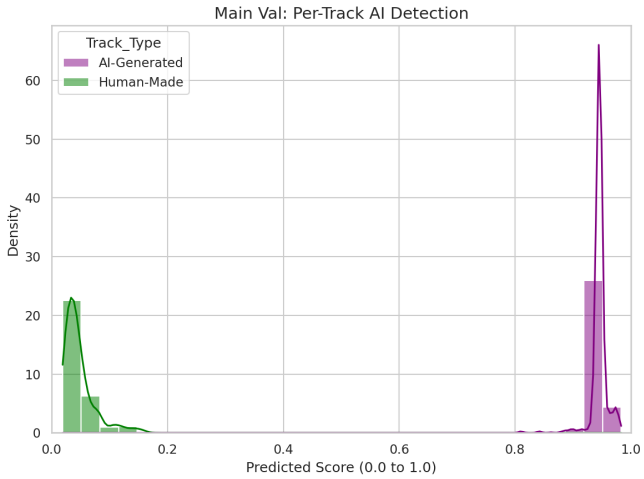
8.3 Score Distribution: Main Validation Set



(a) Similarity score distribution.



(b) Attribution score distribution.



(c) Per-track AI detection score.

Figure 14: Score distributions on the held-out main validation set (693 pairs). Positive (Related) includes any pair with similarity > 0.5 or attribution > 0.5.

The plots confirm the system's learned distinctions:

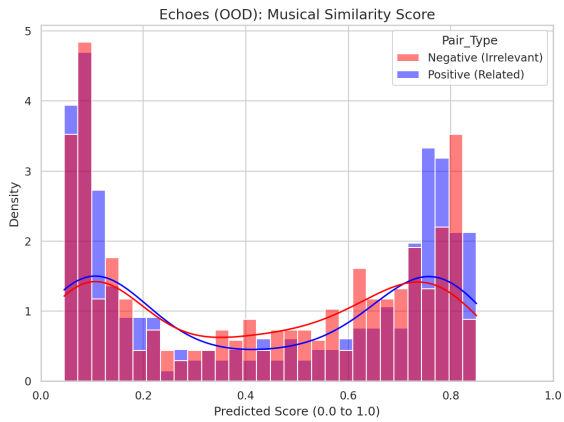
- **Similarity** (Fig. 14a): clean bimodal separation. Negatives concentrate sharply at 0; positives peak at 0.75–0.85 with a spread reflecting varying degrees of musical similarity across lyric pairs, sibling pairs, and MIPPIA covers.

- **Attribution** (Fig. 14b): negatives sharply at ≈ 0.03; positives have a broad distribution with a spike at ≈ 0.97. The flat middle region is *not a failure*—it reflects sibling pairs and MIPPIA covers where the similarity is high but attribution is not always certain. True attribution positives (lyric pairs) drive the spike near 1.0.
- **AI Detection** (Fig. 14c): near-perfect bimodal separation. Human-made tracks are tightly concentrated at 0–0.1; AI-generated tracks at 0.9–1.0. This aligns with the 99.9% AI-A accuracy seen in the ablation.

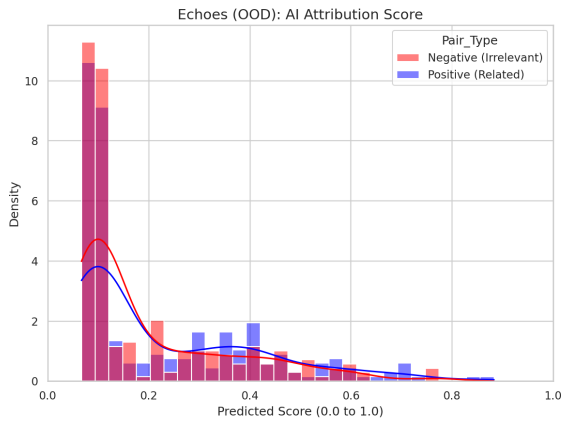
The contrast between the similarity and attribution plots directly illustrates the core design goal: a pair can be "Positive (Related)" with high similarity while still receiving a low attribution score (siblings), and the model correctly distinguishes these cases.

8.4 Out-of-Distribution Evaluation: Echoes

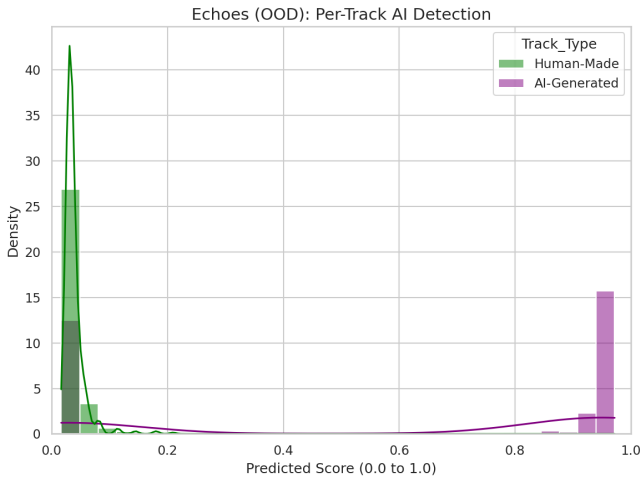
The Echoes dataset was held out entirely from training. It provides a strict OOD benchmark using FMA originals paired with their direct AI derivatives.



(a) Similarity score distribution.



(b) Attribution score distribution.



(c) Per-track AI detection score.

Figure 15: Score distributions on the Echoes OOD dataset. Only Audio-to-Audio (ATA) pairs are included.

The OOD results highlight both the strengths and the remaining limitations of the system:

- **AI Detection generalises well** (Fig. 15c): human-made FMA tracks remain concentrated near 0; Echoes AI derivatives peak at 0.85–0.97. The spread is wider than on main val, reflecting that Echoes generators are unseen during training, but the overall separation is maintained.
- **Similarity separation is weaker OOD** (Fig. 15a): both positive and negative distributions are broadly spread across 0.2–0.8, with only a slight rightward shift for positives. FMA originals are acoustically diverse (jazz, folk, electronic) and the AI derivatives are audio-conditioned, making similarity estimation harder for tracks outside the training distribution.
- **Attribution is conservative OOD** (Fig. 15b): both distributions concentrate near 0, with positives showing a longer but modest tail. The model rarely produces false positive attributions on Echoes, but it also fails to confidently detect many true derivative pairs. This is the primary limitation for out-of-distribution generalization.

The attribution difficulty on Echoes is expected: the model's primary attribution signal comes from lyric pairs (SONICS) and cover pairs (MIPPIA), which are stylistically very different from the audio-conditioned FMA derivatives in Echoes. Improving OOD attribution generalisation would require either including Echoes-like pairs in training (at the cost of OOD integrity) or learning more domain-agnostic attribution features.

9 Limitations

While the multi-task architecture effectively bridges acoustic similarity and synthetic detection, there are several inherent trade-offs:

- **Computational Bottlenecks**: The reliance on large foundation models (MERT-v1-330M and LAION-CLAP) is computationally expensive. Extracting these features requires significant GPU VRAM and processing time.
- **Sensitivity to Mastering Artifacts**: The AI Detection Head relies heavily on "Fakeprint" artifacts and phase deviations. Heavily compressed or aggressively electronic music can occasionally trigger false positives, as extreme digital mastering can mimic the spectral flatness and peak regularity of neural vocoders.
- **Lyric Encoder Coverage**: The Lyric Encoder depends on Whisper transcription quality. Purely instrumental tracks produce zero-vector embeddings, and noisy transcriptions may degrade the lyric signal.

10 Ideas for Future Improvements

- **Segment-Level Attribution Heatmaps**: The `ChunkEncoder` could output a temporal sequence of scores, generating a "heatmap" showing which segments of Track B were derived from Track A.
- **Cross-Attention Multimodal Fusion**: Replacing late fusion with cross-attention between the acoustic `ChunkEncoder` and lyric embeddings, allowing the network to dynamically weigh text similarity during vocal segments and rely on acoustics during instrumental breaks.
- **Advanced Knowledge Distillation**: Scaling the initial training phase by starting with significantly larger foundation models (e.g., MERT-Large or Jukebox-scale representations)

and fine-tuning them directly on the attribution task. This high-capacity "teacher" model can then be distilled into a specialized, task-specific student network that maintains high attribution precision while enabling fast inference.

- **Automated Parameter Optimization**: Utilizing Bayesian optimization or Neural Architecture Search (NAS) to find the optimal balance between the three task-specific losses ($\lambda_{sim}, \lambda_{ai}, \lambda_{attr}$) and regularization parameters. This data-driven approach would replace manual tuning of hyperparameters such as dropout and weight decay, which were found to be critical for preventing overfitting and ensuring cross-domain generalization.
- **Structure-Aware Windowing**: Replacing fixed 10-second windows with beat-aligned segments that respect actual musical bar boundaries [3, 4].

11 Conclusion

This project demonstrates a robust approach to AI-Original pairwise similarity. By shifting the focus from simple binary classification to relational attribution via a Multi-Task Siamese Network, the system effectively quantifies the nuances of musical derivation. The fusion of classical acoustic properties, custom high-order spectral "Fakeprints," deep foundation embeddings, and lyric interaction features ensures that the model can detect synthetic variations even when the generative AI attempts to hallucinate entirely new arrangements.

The exploratory data analysis confirms that AI detection features alone achieve AUC > 0.95, and that fakeprint signatures vary systematically by generator—enabling not only detection but generator-level identification. The decorrelation between classical and fakeprint similarity in SONICS lyric pairs validates the multi-task design: the model must learn to separate "same content" from "same generation process." The strict Out-of-Distribution evaluation on Echoes confirms that the system captures the fundamental signatures of both human musicality and algorithmic synthesis.

12 Tools and Models Used

12.1 Foundation Models

- **MERT** (`m-a-p/MERT-v1-330M`): Self-supervised acoustic model, 1024-d embeddings.
- **LAION-CLAP** (`larger_clap_music`): Contrastive language-audio model, 512-d embeddings.
- **OpenAI Whisper** (Base): Automatic speech recognition for lyric transcription.
- **SBERT** (`all-MiniLM-L6-v2`): Sentence transformer for 384-d lyric embeddings.

12.2 Core Frameworks and Libraries

- **PyTorch & Torchaudio**: Deep learning framework for the Siamese network and audio resampling.
- **Hugging Face Transformers**: Loading and running MERT and CLAP models.
- **Librosa**: Classical acoustic feature extraction (MFCCs, Mel-spectrograms, Chroma, Spectral Contrast).
- **SciPy & Scikit-learn**: Statistical validation and dimensionality reduction in EDA.
- **Pandas & NumPy**: Metadata management, caching, and vectorized operations.
- **Matplotlib & Seaborn**: Visualisation of feature distributions and evaluation plots.

12.3 Environment

Python 3.13, managed via `conda`. All audio standardised to 16 kHz, mono, PCM_16 WAV.

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